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LLM Use in Early Prototyping

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Abstract

Early-stage prototyping is crucial in software design and recent advances in LLMs are reshaping this phase. LLM-based tools bridge gaps between requirements and design, allowing faster iterations and more integrated workflows. Prompt engineering lets multidisciplinary teams to rapidly explore design ideas and assess AI capabilities before full-scale development. This paper shifts authors' previous pedagogical thesis toward a technical focus; examining the role of LLMs in the early prototyping of an accessible application. The case study, NeuRoam, is a tablet-based educational app prototype for neurodivergent children, aligned with WCAG 2.2 and COGA guidelines. Analysis covers how LLM tools accelerate the prototyping of such inclusive apps while ensuring adherence to accessibility and user requirements. Results from a structured evaluation using 17 LLMs showed that models with stronger reasoning capabilities significantly predicted higher usability ratings ($\beta \approx 0.823$, $p = 0.002$), with the regression model explaining approximately 55% of score variance ($R^2 \approx 0.555$). LLM feedback identified both critical accessibility issues (e.g., low contrast) and praised features such as gamification, highlighting their potential as early-stage evaluators in inclusive design.

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1. Introduction

In educational technology design, early-stage prototyping ensures diverse learner needs are addressed from the outset. For neurodivergent children (e.g., with autism, ADHD, or dyslexia), designing for accessibility early on can improve engagement and outcomes.

However, implementing standards such as WCAG 2.2 in practice remains challenging, often requiring specialized expertise. Therefore, they are rarely fully implemented and leaving many apps not adequately designed for learners with cognitive and learning differences. As a result, these children often encounter obstacles such as sensory overload from cluttered interfaces, ambiguous instructions, or unpredictable interface behavior that disrupts engagement. Designing prototypes that address such challenges and fully comply with accessibility specifications is a complex, resource intensive-endeavor requiring specialized expertise.

Recent advances in artificial intelligence offer a promising remedy for this bottleneck. Large language models have shown potential as tools to accelerate early prototyping while keeping inclusion in focus. By adopting LLMs during the initial design phase, developers can generate accessible interface content, simplify language and even simulate user feedback. This approach reduces the time and expertise needed to iterate on designs, ensuring that key inclusive design principles are embedded from the outset. In essence, LLM can serve both as creative assistants and evaluators, helping translate abstract accessibility guidelines into practical design improvements during prototyping.

This paper builds upon earlier pedagogical research [1] in inclusive education and shifts toward more technical exploration. Previous studies underscored the significance of universal design principles in engaging neurodivergent learners. Building on those insights, this study investigates how different LLMs influence the evaluation of the NeuRoam prototype. The primary interest lies in determining how effectively LLMs perform in supporting decision-making processes during the early stages of prototyping accessible educational tools.

As a case study, NeuRoam, a tablet based educational application was designed for neurodivergent children. NeuRoam was developed to conform to WCAG 2.2 and COGA guidelines, providing interactive learning activities through an accessible interface. This case study illustrates the potential benefits of integrating LLM support into early prototyping to achieve inclusive design goals more efficiently. Next, the background and related work that frames this approach within the broader landscape of educational technology and AI-assisted design is reviewed.

2. Related work

In line with authors' prior work developing an accessible educational app for neurodivergent children, guided by the standards, recent research emphasizes aligning AI innovations with inclusive design principles. For example, Korada et al. [2] present a conceptual framework that synthesizes AI with established accessibility guidelines, stressing that AI-driven systems must be adaptable to diverse user needs and uphold ethical standards like privacy and informed consent. Complementing this view, Ferreira and Castro [3] report that fewer than 2% of studies (2013-2023) provide a systematic review of participatory and inclusive design models for children with autism. Screening over two genuinely embedded autistic children under Universal Design (UD) principles. Their conclusions reinforce the aim of using LLM support during early prototyping so that accessibility heuristics shape interfaces from the first wireframe.

Building on this foundation, several studies have started probing how LLM-based tools could directly support learners in a special education context. Ronksley-Pavia et al. [4] conducted a scoping review of generative AI applications for neurodivergent students. They identified promising use cases such as AI-driven personalized learning aids and automated drafting of individualized education plans that could help students with ADHD, autism, dyslexia, and other conditions. Notably, the review found generative AI could provide real-time, personalized support for learners while reducing administrative burdens on educators. However, it also raised cautions: the evidence base is still very limited (only 9 empirical studies) and experts worry about accuracy over-reliance on AI, privacy and ultimately, the necessity of human oversight. These concerns are highlighted in domain-specific investigations as well. For instance, a recent study explored ChatGPT's potential to enhance therapy for children with ADHD. Using a Delphi method with child therapy experts, Berrezueta-Guzman et al. [5] simulated an ADHD counseling session with a custom ChatGPT and found the model demonstrated strengths like empathy, adaptability and effective communication. The experts agreed such an LLM could help personalize and scale ADHD interventions, pointing to a "new era" of AI-assisted therapy. Although GPT-4 is emerging as a generally accurate, empathic support tool for

ADHD and autism care, its clinical value is still limited by privacy concerns, cultural sensitivity gaps, weak handling of non-verbal cues, and a shortage of actionable, fully reliable guidance [6].

Parallel efforts are bringing LLM assistance into the software prototyping and development workflow to proactively embed accessibility. Cali et al. [7] introduce a prototype Visual Studio Code extension that uses LLM calls to help developers identify and fix web accessibility issues during coding. This tool, integrated into the IDE, checks code against accessibility guidelines and suggests fixes in real time, aiming to catch issues before they reach production. Beyond software tooling, generative AI is also explored as a content creation partner in education. Leiker et al. [8] report on a prototype pipeline that uses an LLM to generate adult learning course materials at scale. Their system accelerated the production of training modules by having the LLM draft lessons and assessments, which were then reviewed and refined by experts. Initial results on both studies showed that strategic and pedagogical decisions should remain under human oversight as the detection accuracy is uneven and can lead to some missed or false-positive issue identifications.

Concurrently, a wave of recent work has investigated how LLMs can serve as design *partners* in early-stage UI/UX prototyping. This spans from helping generate and evaluate interface designs to enabling new collaborative workflows between designers and AI. Specifically, Duan et al. [9] developed a Figma plugin that lets GPT-4 perform heuristic reviews of UI mock-ups, catching spacing inconsistencies, unclear labeling, and other subtle issues that human designers often overlook. Its feedback accelerates early refinement, but adds little value in later iterations, so designers should treat the model as an initial “junior reviewer” rather than a substitute for expert judgment. Similarly, Petridis et al. [10] introduced PromptInfuser – a Figma plugin that allows designers to embed LLM prompts directly into UI mockups. PromptInfuser allows designers to bind to GPT-4 prompts directly to Figma mock-up elements so that UI and AI logic evolve together in semi-functional prototypes. In tests with several UX designers, this sped up iterations and produced more realistic prototypes while surfacing technical constraints like API limits and latency. Another “live prototyping” with generative AI idea is proposed by Santiago et al. [11], letting designers alter AI outputs and UI elements in real time during user tests via a control panel, so the feedback is integrated immediately. This speeds insight generation and iteration but requires greater AI expertise and contingency planning from the design team.

There is also progress in using LLMs and multimodal models to automatically generate UI layouts and visuals from high-level specification, which can accelerate the early prototyping of interfaces. Lin et al.’s LayoutPrompter [12] shows that with only structured in-context prompts and a ranking step, a *vanilla* LLM can generate GUI layouts that meet or beat state-of-the-art specialist models, even in low-data settings. The work demonstrates that LLMs can serve as versatile “design engines” while handling text-to-layout and constraint-based tasks through prompt engineering alone.

All papers considered and discussed above have collectively advanced and sharpened interest in a related yet distinct line of inquiry – evaluating a prototype once it has already been produced by a human designer. This is precisely where authors work begins, leading into the more detailed research presented in the next section.

3. Methods

3.1. Problem Definition and Objectives

This study is focused on the use of LLMs as evaluation tools during early prototyping of the accessible educational application NeuRoam. The primary objective is to explore how feedback generated by LLMs can accelerate design iterations while ensuring adherence to WCAG 2.2 and COGA accessibility guidelines.

Residual diagnostics confirmed acceptable model assumptions: residuals appeared roughly normal, as visualized in the QQ plot seen in Figure 1, while variance inflation factors ($VIF \approx 1.36$) indicated low multicollinearity. In addition, to assess presence of heteroscedasticity, the Breusch-Pagan test was conducted and showed no significant evidence ($p = 0.129$).

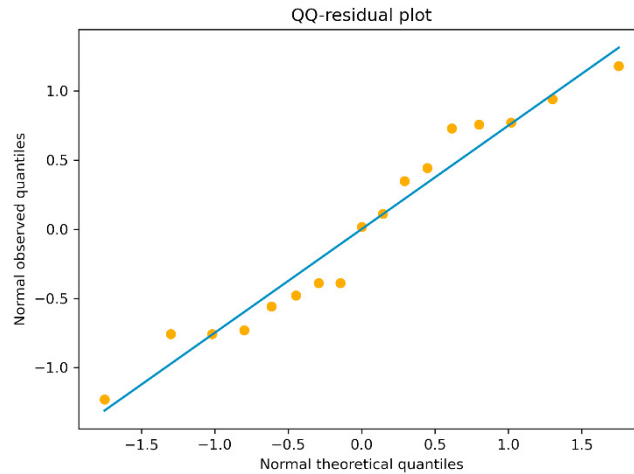


Figure 1. QQ-residual Plot

3.2. Synthetic Evaluation Approach

A synthetic user evaluation approach is adopted; meaning that LLMs are used as artificial evaluators that provide usability and accessibility on high-fidelity prototype. This method is grounded in evidence that LLMs (e.g., GPT-4.1) effectively apply design heuristics and accessibility guidelines to produce useful early-stage feedback when integrated into the design process. Here, each LLM acts as a “synthetic evaluator”, highlighting the potential design flaws or accessibility issues, and with awareness of inherent limitations such as occasional inaccuracies or hallucinations. The selected models were categorized geographically into three regions: America, Europe and Asia, reflecting diverse origins of development. An overview of all 17 evaluated models, along with their year attributes and regional categorization, is provided in Table 1. The overall model fit was moderate ($R^2 \approx 0.555$). Among predictors, reasoning ability showed a strong standardized effect ($\beta = 0.823, p = 0.002$).

Table 1. LLM detailed overview

LLM	Year	Ownership
ChatGPT (3, o3, 04-mini, o4-mini-high, 4o, 4.1, 4.5)	2022-2025	OpenAI, America
Gemini (1.5 Flash 8B, 2.5 Flash)	2024-2025	Google DeepMind, America
Gemma 3 (4G, 12B, 27B)	2025	Google DeepMind, America
Claude Sonnet 4	2025	Anthropic, America
Perplexity PRO	2024	Perplexity AI, America
Grok3	2024	xAI, America
Mistral 7B	2023	Mistral AI, Europe
DeepSeek R1	2025	DeepSeek, Asia

Although three hypotheses were tested (H1-H3), no formal correction for multiple comparisons was applied due to the exploratory nature of the analysis.

3.3. Prompt-based Evaluation Procedure

The evaluation was conducted by developing a standardized series of prompts simulating seven key interactions (user login, navigation, task completion, etc.) with the NeuRoam prototype seen on Figure 2.

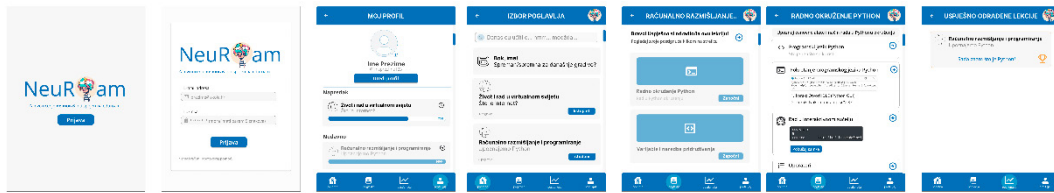


Figure 2. NeuRoam - AiO prototype (Figma)

All 17 selected LLMs received identical prompts to ensure consistency, instructing them to rate usability on a 1-5 scale and provide explanatory feedback for each interaction. This structured approach simulated traditional user evaluations and allowed precise identification of strengths and weaknesses in accessibility compliance (e.g., clarity of instruction, contrast requirements).

The evaluation specifically tested three hypotheses:

- H1: More advanced (e.g., multimodal input, reasoning option) LLMs will provide better ratings
- H2: Newer generation LLMs will pinpoint possible mistakes more accurately
- H3: LLMs developed in the United States will *judge* prototypes more leniently

Whereas the system role is defined as: “*Imagine you are a neurodivergent child – more precisely, a fifth grader with ADHD. In front of you is a prototype app interface NeuRoam that should make learning easier. Today you’ve chosen to study computer science, focusing on the lesson “Python: Working Environment”. Keep in mind that you know the WCAG and COGA guidelines so judge the prototype accordingly. Try to keep your comments brief; meaning you should avoid sharing too much detail but write them in standard literary Croatian. At the end of each step, give a rating from 1 to 4. You’ll receive the prototype once you confirm you’re ready!*”

3.4. Data Collection and Analysis

Data collection involved quantitative usability ratings and qualitative explanations provided by all 17 distinct LLMs, covering different model generations, capabilities, and origins. Quantitative analysis was performed using multiple linear regression to statistically test hypotheses H1-H3 and explore model differences with independent variables: *multi_reasoning*, *year*, *ownership* – three predictors that correspond to the stated hypotheses. An average prototype mark was then computed (labelled *mark-avg*), which served as the dependent variable in the conducted statistical tests. Furthermore, qualitative data underwent thematic analysis to identify recurring accessibility and usability issues mentioned by the models, providing deeper insight into the evaluation consistency. This will be examined in more details through *Qualitative analysis* within *Results* section of this paper.

4. Results and Discussion

Given the limited availability of high-performing models with known capabilities, a total of 17 distinct LLMs were selected. While this sample provides adequate variation across the studied predictors, it represents a statistical constraint. Therefore, results from regression analysis are interpreted cautiously.

4.1. Quantitative Analysis

Multiple regression analysis of the LLM-simulated evaluations indicated a moderate model fit ($R^2 \approx 0.555$), suggesting that the three selected LLM features explained roughly 55% of the variance in the prototype’s average usability scores. Specifically, the full model explained 55.5% of the variance in usability ratings. The semantic reasoning capability of an LLM was a strong positive predictor of the rating ($\beta \approx 0.823$, $p = 0.002$), implying that more advanced models consistently gave higher usability scores to the NeuRoam prototype. In contrast, neither the

model's origin nor its release year had a significant effect on the scores ($p = 0.266$ and 0.949 , respectively), so statistically only the first hypothesis was confirmed, while H2 and H3 remained unsupported by the quantitative data.

Regression diagnostics, seen in Table 2, indicated no major assumption violations: all variance inflation factors were low (≈ 1.36), suggesting negligible multicollinearity among predictors. The residuals were approximately normally distributed aside from minor tail deviations (as expected given the small sample, $n = 17$), and a Breusch-Pagan test showed no evidence of heteroscedasticity.

Table 2. Key statistical metrics

Parameter	Expected value	Result	Validation
R^2	> 0.1	0.555	✓
VIF (variance inflation factor)	< 3.3	H1: 1.357 H2: 1.357 H3: 1.362	H1: ✓ H2: ✓ H3: ✓
p	< 0.05	H1: 0.002 H2: 0.206 H3: 0.949	H1: ✓ H2: X H3: X

Extending the model with moderator variables revealed that the positive effect of reasoning ability on usability ratings was significantly amplified for proprietary (closed-source) LLMs compared to open-source ones (interaction $p = 0.039$) as seen in Figure 3. By contrast, model release did not significantly moderate this relationship, indicating that generational differences in LLMs did not alter the influence of reasoning capabilities on evaluation outcomes.

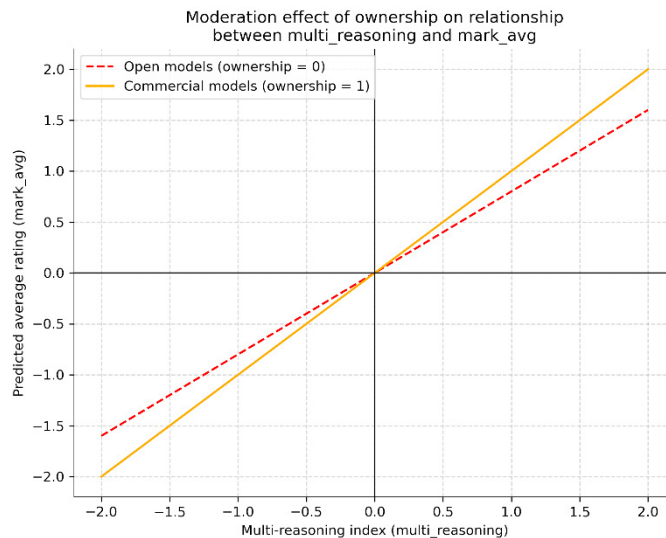


Figure 3. Moderation effect of ownership

4.2. Qualitative Feedback Summary

Qualitative analysis of the LLM's feedback complemented these results and provided actionable design insights. Across the six core usability tasks, average LLM-given scores ranged from ~ 3.4 to ~ 4.2 out of 5, reflecting generally moderate to high satisfaction with the prototype's accessibility and usability. LLMs flagged various accessibility and UX issues aligned with WCAG 2.2 and COGA recommendations; for instance, they noted the (purposely left out)

absence of a “Show Password” option and poor text contrast on the login screen (violations of WCAG Form Input and Contrast criteria). They also critiqued the use of long, unsegmented text blocks in the learning content, pointing out that this lack of content chunking could impede cognitive processing for neurodivergent users.

Later in the user flow, LLMs’ feedback was more positive – e.g., they praised the profile page’s clear progress visualization (progress bars and percentages) as an effective motivational element and highlighted the achievements screen’s trophy icon as a simple but powerful gamification feature that encourage continued learning. Interestingly, U.S.-based models tended to deliver their critiques in a more polite and encouraging tone despite similar numeric ratings, suggesting a cultural difference in feedback style that aligns qualitatively with the hypothesized leniency of American LLMs (H3).

4.3. Interpretation and Broader Implications

These results suggest that while LLMs are promising evaluators during prototyping, their feedback is shaped by the model’s underlying reasoning ability more than its origin or recency. This raises open questions for the design of AI-assisted workflows: how to balance model strengths with human oversight, and how to ensure culturally neutral, context-sensitive design advice. As this study focused on accessibility in education, future research should examine how well these synthetic insights transfer to other domains with similar inclusion challenges.

4.4. Ethical Considerations

This study involved no human subjects, so formal ethics was not required. All LLM queries were conducted via publicly available APIs, using only model-generated content and researcher-designed prototype. No personal or user data were collected at any point. While privacy risks are minimal, energy costs and licensing constraints are valid concerns in deploying propriety LLMs in classroom contexts. These issues warrant further discussion as part of future implementation studies.

5. Limitations

Using LLM-based synthetic evaluation is innovative and scalable, but it has clear limits. Most importantly, the study does not include feedback from actual end users; namely, the neurodivergent learners the prototype targets. Without their input, the results, though semantically rich, may miss subtle, real-world usability issues that surface only in live use. Additionally, while the hypotheses (H1–H3) rest on plausible assumptions (e.g., the importance of reasoning ability and model maturity), their theoretical grounding would benefit from a deeper review of the literature on LLM behavior and evaluation patterns. Another limitation concerns the relatively small sample size ($n = 17$), which restricts the statistical power of the regression analysis. While the findings are indicative and supported by diagnostics, they should be interpreted with caution. Furthermore, this study focuses on a single prototype at an early development stage and does not include baseline comparison with expert human evaluations.

As such, its contribution is necessarily scoped and demonstrates how LLMs can support accessibility-focused prototyping, particularly for WCAG 2.2 and COGA compliance in neurodivergent contexts. Future research should aim to validate these results with a broader set of models and complement synthetic evaluations with real-user data to strengthen generalization.

6. Conclusion

Overall, key findings confirm that LLMs are rapidly becoming enablers in the UI/UX design process: from automating routine evaluation and generation tasks to facilitating new prototyping workflows that integrate AI’s generative power with human creativity. Their ability to instantly evaluate UI variants, suggest compliant copy and even generate production-ready snippets, enables teams to explore exponentially more ideas. However, as indicated by quantitative results from multiple regression, technical speed alone is insufficient: prototypes incorporating accessibility criteria from the earliest stage were found significantly more usable and widely accepted by stakeholders.

Upholding compliance of AI-generated outputs with WCAG and COGA standards is therefore recognized as both an ethical imperative and a practical accelerator of user satisfaction, particularly for learners experiencing cognitive or sensory difficulties. Realizing this potential will require curricular changes and systematic professional training, ensuring accessibility is viewed as a foundational principle rather than an afterthought. Thus, the formation of interdisciplinary teams is advocated, allowing AI-augmented tools to be continuously audited and refined based on authentic classroom feedback. In short, while prototyping processes can be accelerated by LLM technology, meaningful progress can only be achieved through an inclusive approach.

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